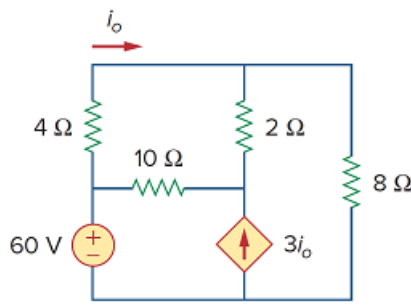


Problem

Using nodal analysis, find current i_o in the circuit of Fig. 3.66.

Figure 3.66:



Step-by-step solution

Step 1 of 5

Refer to the circuit diagram depicted in Figure 3.66 in the textbook.

Re-draw modified circuit diagram notating with node voltages v_1, v_2 and v_3 .

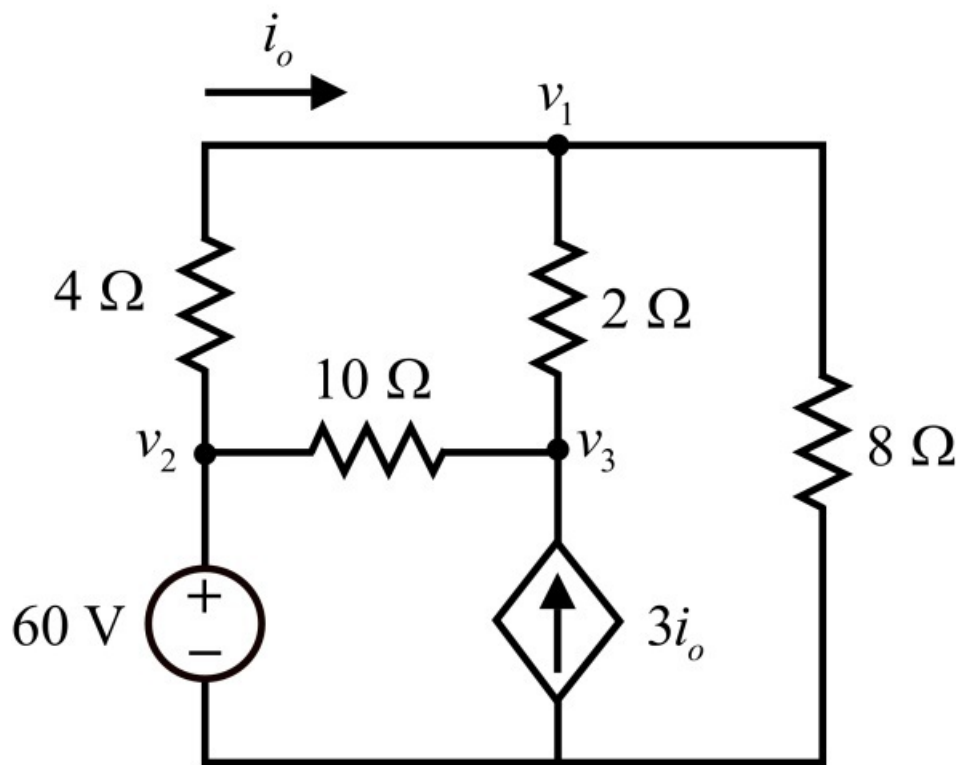


Figure 1

Step 2 of 5

From Figure 1, it is observed that,

The voltage at node 2 is,

$$v_2 = 60 \text{ V}$$

The value of current i_o is,

$$i_o = \frac{v_2 - v_1}{4}$$

$$i_o = \frac{60 - v_1}{4} \dots\dots (1)$$

Step 3 of 5

Apply Kirchhoff's current law at node v_1 .

$$\frac{v_1 - v_2}{4} + \frac{v_1 - v_3}{2} + \frac{v_1}{8} = 0$$

$$\frac{v_1 - 60}{4} + \frac{v_1 - v_3}{2} + \frac{v_1}{8} = 0$$

$$\frac{2v_1 - 120 + 4v_1 - 4v_3 + v_1}{8} = 0$$

$$7v_1 - 4v_3 = 120$$

$$4v_3 = 7v_1 - 120$$

$$v_3 = \frac{7v_1 - 120}{4} \dots\dots (2)$$

Apply Kirchhoff's current law at node v_3 .

$$\frac{v_3 - v_2}{10} + \frac{v_3 - v_1}{2} - 3i_o = 0$$

$$\frac{v_3 - 60}{10} + \frac{v_3 - v_1}{2} - 3\left(\frac{60 - v_1}{4}\right) = 0 \quad \left(\text{since } i_o = \frac{60 - v_1}{4}\right)$$

$$\frac{v_3 - 60}{10} + \frac{v_3 - v_1}{2} - \frac{180 - 3v_1}{4} = 0$$

$$\frac{2v_3 - 120 + 10v_3 - 10v_1 - 900 + 15v_1}{20} = 0$$

$$5v_1 + 12v_3 = 1020 \dots\dots (3)$$

Step 4 of 5

Substitute $\frac{7v_1 - 120}{4}$ for v_3 in equation (3).

$$5v_1 + 12\left(\frac{7v_1 - 120}{4}\right) = 1020$$

$$5v_1 + 21v_1 - 360 = 1020$$

$$26v_1 = 1380$$

$$v_1 = 53.08 \text{ V}$$

Therefore, the voltage at node v_1 is 53.08 V .

Step 5 of 5

Determine the current i_o using equation (1).

$$\begin{aligned} i_o &= \frac{60 - v_1}{4} \\ &= \frac{60 - 53.08}{4} \\ &= 1.73 \text{ A} \end{aligned}$$

Therefore, the value of current i_o in the circuit is 1.73 A.